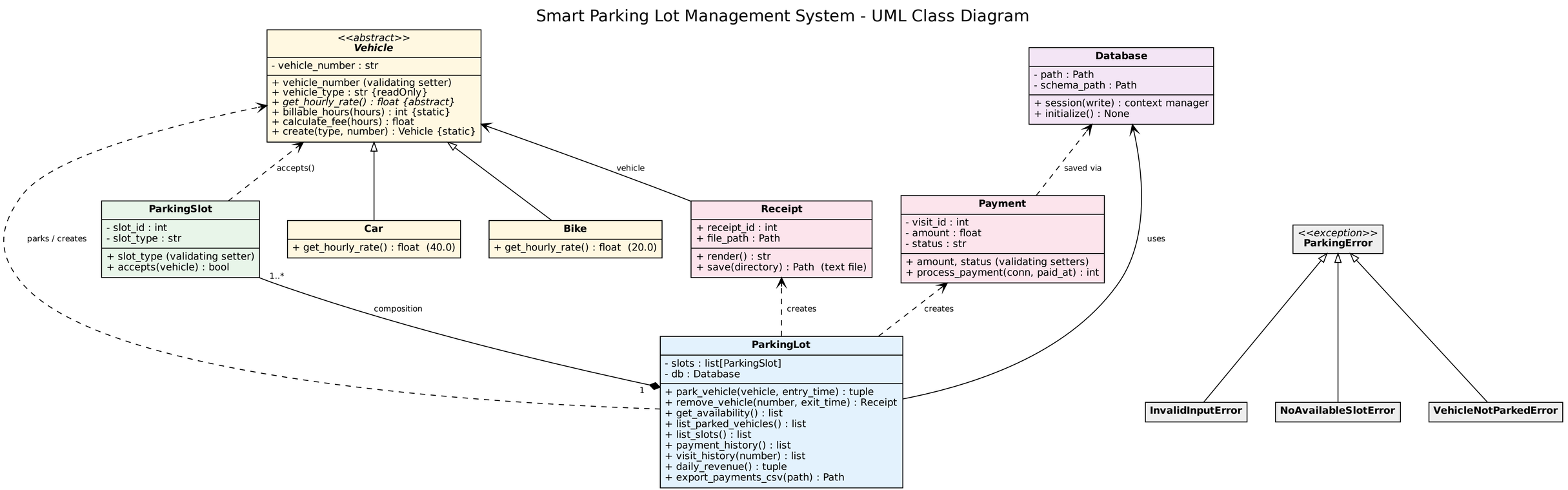
Smart Parking Lot Management System - UML Class Diagram
<<abstract>>
Vehicle
- vehicle_number : str
+ vehicle_number (validating setter)
+ vehicle_type : str {readOnly}
+ get_hourly_rate() : float {abstract}
+ billable_hours(hours) : int {static}
+ calculate_fee(hours) : float
+ create(type, number) : Vehicle {static}
Database
- path : Path
- schema_path : Path
+ session(write) : context manager
+ initialize() : None
ParkingSlot
- slot_id : int
- slot_type : str
+ slot_type (validating setter)
+ accepts(vehicle) : bool
Car
+ get_hourly_rate() : float (40.0)
Bike
+ get_hourly_rate() : float (20.0)
Receipt
+ receipt_id : int
+ file_path : Path
+ render() : str
+ save(directory) : Path (text file)
Payment
- visit_id : int
- amount : float
- status : str
+ amount, status (validating setters)
+ process_payment(conn, paid_at) : int
<<exception>>
ParkingError
InvalidInputError
NoAvailableSlotError
VehicleNotParkedError
ParkingLot
- slots : list[ParkingSlot]
- db : Database
+ park_vehicle(vehicle, entry_time) : tuple
+ remove_vehicle(number, exit_time) : Receipt
+ get_availability() : list
+ list_parked_vehicles() : list
+ list_slots() : list
+ payment_history() : list
+ visit_history(number) : list
+ daily_revenue() : tuple
+ export_payments_csv(path) : Path
accepts()
vehicle
saved via
parks / creates
1..*
composition
1
creates
creates
uses